

RYAN DSOUZA

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SUMMARY

Data Scientist with experience building forecasting models, GenAI automation pipelines, and full-stack analytics tools in financial services; M.S. Applied Data Science candidate (4.0 GPA).

WORK EXPERIENCE

KPMG — Chicago, IL | Advisory Associate, Data Scientist

Aug 2025 – Present

Revenue Forecasting | Internal

- Developed 3 time series forecasting models (BSTS, SARIMAX) across Advisory, Tax, and Audit lines of business, integrating Moody's macroeconomic signals to achieve 87% forecast accuracy.
- Enabled scenario-based revenue projections under conditions such as government shutdowns and market downturns, validated through sensitivity analysis to optimize variable selection.

Data Reconciliation & Platform Migration | Financial Services Client

- Engineered a Python-based reconciliation pipeline (multiprocessing) to process 10M+ rows of banking data, applying bank-sweep, account-hierarchy, and security-ID logic to capture 80% of discrepancies and identify 10–400+ mismatches per bank daily.
- Transitioned reconciliation workflow to a SQL-driven pipeline feeding a Power BI dashboard for real-time stakeholder monitoring, reducing project timeline from 4 months to 3 weeks. (SageMaker, SQL, Python, Power BI)

Control Testing Automation | Financial Services Client

- Architected a modular Snowflake data pipeline (12 Python scripts) processing historical and current-period control data in separate streams before joining to minimize query runtime, encapsulating completion, compliance-testing, and LOB-mapping logic into a validated HTML compliance report — designed for incremental migration from Python to simplified SQL as the system of record matures.
- Built a Claude-powered agent pipeline guided by a custom SKILL.md to auto-generate narrative testing-outcome summaries from validated SQL-extracted results across 13 business units — eliminating a month of manual testing time and reducing effort equivalent to 2 FTEs per reporting cycle.
- Automated a Google Sheets dashboard (Apps Script, JavaScript, SQL) tracking 1,200+ controls across 13 business units and 7 test leads, cutting reporting turnaround from 24 hours to 1 hour.

Audit Automation Platform | Internal

- Built and trained 3 NLP models to score audit controls across 4 dimensions, achieving 78% precision@5 against auditor-sourced labels, aggregated into an automation feasibility rating across 7,600+ controls.
- Developed a full-stack web app (Streamlit, Flask, Celery, Redis) used by 12 auditors, cutting manual audit review from 3–4 hours to minutes via a real-time dashboard and batch LLM recommendation pipeline.

KPMG — Chicago, IL | Data Science Intern

Jun 2023 – Aug 2024

- Supported data migration for 5 business lines, mapping 10,000+ rows at 95% accuracy for a smooth system transition.
- Collaborated cross-functionally to design scalable data architecture, boosting system performance by 20%.
- Led COBOL modernization research, proposing an AI/ML roadmap projected to cut operational costs and technical debt by 30% over 5 years.

DePaul University — Chicago, IL | Data Analyst & Transfer Peer Mentor

Aug 2024 – Aug 2025

- Built automated Python (Pandas, NumPy) reporting scripts to track 150+ mentees' progress, reducing manual data collection time by 20%.
- Analyzed mentee engagement trends in Jupyter Notebook, surfacing success drivers that helped leadership set clearer mentor expectations and tailor guidance strategies.

SKILLS

Languages: Python, SQL, R, Java, JavaScript, PySpark

ML / GenAI: Scikit-learn, TensorFlow, XGBoost, Pandas, NumPy, Claude/LLM Agents, Prompt Engineering

Cloud & Infra: AWS (SageMaker, Athena, EMR), GCP, Docker, Git/GitHub, Databricks, Snowflake

Visualization: Matplotlib, Seaborn, Plotly, Tableau, Power BI

Other: Flask, Streamlit, Celery, Redis, A/B Testing, Agile

CERTIFICATIONS

AWS Certified Cloud Practitioner — AWS (Oct 2025–Oct 2028) | AWS Certified Data Engineer Associate — Exam Prep, Udemy (Aug 2025) | Data Engineering Foundations — LinkedIn Learning (Jun 2024)

EDUCATION

University of Chicago — M.S. in Applied Data Science (GPA: 4.0/4.0)

Expected Dec 2026

Coursework: ML Ops, Big Data Engineering, Computer Vision, Advanced Machine Learning, Time Series Analysis, Optimization & Simulation Methods for Data Science

DePaul University — B.S. in Data Science & Mathematics (GPA: 3.98/4.0)

Jun 2025

Summa Cum Laude